1(a)	$(f(1)=)1-5+e=-1.281...<0$ $(f(2)=)2^2-5\times2+e^2=1.389...>0$	M1
	As there is a change of sign and $f(x)$ is continuous over the interval $[1, 2]$ then there is a root *	A1*
		(2)
(b)(i)	$x_2=\sqrt{5\times1-e^1}=$ awrt 1.5105	M1A1
(ii)	$\alpha=$ awrt 1.7340	A1
		(3)
		(5 marks)

2(a)	$\text{ff}(6) = \text{f}\left("\frac{9}{2}"\right) = \frac{"9/2"+3}{"9/2"-4} = ...; \ = 15$	M1; A1
		(2)
(b)	$\text{f}^{-1}(x) = \frac{4x+3}{x-1}$	M1A1
	$x \in \mathbb{R},\ x \neq 1$	B1
		(3)
(c)	E.g. $\left(\frac{x+3}{x-4}\right)^2 + 5 = 7$ or $\frac{a+3}{a-4} = (\pm)\sqrt{7-5}$	M1
	$\Rightarrow x^2 - 22x + 23 = 0 \Rightarrow x = ...$ or $(a+3) = (\pm)\sqrt{2}(a-4) \Rightarrow a = ...$	dM1
	$(a =) 11 + 7\sqrt{2}$ oe	A1
		(3)
		(8 marks)

3(a)	$(\cos 2A \equiv)\cos A\cos A - \sin A\sin A \equiv \cos^2 A - (1-\cos^2 A)$	M1
	$\Rightarrow \cos 2A \equiv 2\cos^2 A - 1$ *	A1*
		(2)
(b)	$\int \left(3-2\cos 6x\right)\mathrm{d}x = 3x - \frac{\sin 6x}{3} \quad (+c)$	M1A1
	$\left[3x - \frac{\sin 6x}{3}\right]_{\frac{\pi}{12}}^{\frac{\pi}{8}} = \left(3\left(\frac{\pi}{8}\right) - \frac{\sin\left(6\times\frac{\pi}{8}\right)}{3}\right) - \left(3\left(\frac{\pi}{12}\right) - \frac{\sin\left(6\times\frac{\pi}{12}\right)}{3}\right) = \frac{1}{8}\pi + \frac{2-\sqrt{2}}{6}$	dM1A1
		(4)
		(6 marks)

4(a)	$A = 93$	B1
		(1)
(b)	$100 = 125 - "93"\mathrm{e}^{-0.109T} \Rightarrow A\mathrm{e}^{-0.109T} = ...$	M1
	$A\mathrm{e}^{kT} = B \Rightarrow T = \dfrac{\ln\left(\dfrac{B}{A}\right)}{k}$	dM1
	$T = 12.05$	A1
		(3)
(c)	$\dfrac{\mathrm{d}N}{\mathrm{d}t} = 0.109 \times "93"\mathrm{e}^{-0.109\times 7}$	M1
	4 730 (total sales per month)	A1
		(2)
(d)	The limit is 125 000 / the model has a limit below 150 000	B1
		(1)
		(7 marks)

5(a)	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{(x^2+k)\times\dfrac{2x}{x^2+k} - 2x\ln(x^2+k)}{(x^2+k)^2}$	M1
	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2x-2x\ln(x^2+k)}{(x^2+k)^2} = \dfrac{2x(1-\ln(x^2+k))}{(x^2+k)^2}$	M1A1
		(3)
(b)	$x=0$	B1
	$"1"-\ln(x^2+k)=0 \Rightarrow x^2 = \mathrm{e}^{"1"} \pm k$	M1
	$x=\pm\sqrt{\mathrm{e}-k}$	A1ft
		(3)
(c)	Upper limit is e or $k < \mathrm{e}$	B1ft
		(1)
		(7 marks)

6(a)	$\log_{10} S = 4.5 - 0.006 \times 2 \Rightarrow S = 10^{4.5-0.006\times 2} = 30800\,\text{km}^2$	M1A1
		(2)
(b)	$\log_{10} S = 4.5 - 0.006t \Rightarrow S = 10^{4.5-0.006t}$ (or $p = 10^{4.5}$ or $q = 10^{-0.006}$)	M1
	$S = 10^{4.5-0.006t} = 10^{4.5} \times \left(10^{-0.006}\right)^t$ (or $p = 10^{4.5}$ and $q = 10^{-0.006}$)	dM1
	$S = 31600 \times (0.986)^t$	A1
		(3)
(c)	E.g. The proportion of area covered by coral reefs retained from year to year.	B1
		(1)
		(6 marks)

7(a)	$\mathrm{f}(0)=(0-3)^2=9$	M1
	$0 \leqslant \mathrm{f}(x) \leqslant 9$	A1
		(2)
(b)	$\mathrm{f}'(x)=-2x\mathrm{e}^{-x^2}(2x^2-3)^2+\mathrm{e}^{-x^2}\times 8x(2x^2-3)$	M1A1
	$=2x(2x^2-3)\mathrm{e}^{-x^2}\left(-(2x^2-3)+4\right)=2x\mathrm{e}^{-x^2}(2x^2-3)\left(7-2x^2\right)$	dM1A1
		(4)
(c)	$x^2=\frac{3}{2},\frac{7}{2}\Rightarrow \mathrm{f}\left(\sqrt{\frac{7}{2}}\right)=\mathrm{e}^{-\frac{7}{2}}\left(2\times\frac{7}{2}-3\right)^2=16\mathrm{e}^{-\frac{7}{2}}$	M1A1
	$16\mathrm{e}^{-\frac{7}{2}}<k<9$	dM1A1
		(4)
		(10 marks)

8(a)	Starting with the LHS: $2\text{cosec}^2 2\theta(1-\cos 2\theta) = \dfrac{2-2\cos 2\theta}{\sin^2 2\theta}$	M1
	$= \dfrac{2-2(1-2\sin^2\theta)}{4\sin^2\theta\cos^2\theta}$	M1dM1
	$= \sec^2\theta = 1+\tan^2\theta \equiv \text{RHS}$ *****	A1*
		(4)
(b)	$\sec^2 x - 3\sec x - 4 = 0 \Rightarrow \sec x = ...$	M1
	$\cos x = \dfrac{1}{4}$ (ignore -1)	A1
	$\cos x = \dfrac{1}{4} \Rightarrow x = ...$	dM1
	$x = 75.5°,\ 284.5°$	A1
		(4)
		(8 marks)

(b)Alt	$1+\tan^2 x = 4+3\sec x \Rightarrow 9\sec^2 x = \tan^4 x - 6\tan^2 x + 9 \Rightarrow \tan^4 x - 15\tan^2 x = 0$ $\Rightarrow \tan^2 x = ...(\neq 0)$	M1
	$\tan x = (\pm)\sqrt{15}$	A1
	$\tan x = k \neq 0 \Rightarrow x = ...$	dM1
	$x = 75.5°,\ 284.5°$	A1
		(4)

9(a)	$k = -1$	B1
		(1)
(b)(i)	$\mathrm{f}(0) = \lvert 2 - 4\ln(0+1) \rvert = 2 - 0 = 2$	B1
(ii)	$0 = 2 - 4\ln(x+1) \Rightarrow \ln(x+1) = \frac{1}{2} \Rightarrow x = \mathrm{e}^{p} + q$	M1
	$x = \mathrm{e}^{\frac{1}{2}} - 1$	A1
		(3)
(c)	$2 - 4\ln(x+1) = 3 \Rightarrow \ln(x+1) = \ldots$ or $-2 + 4\ln(x+1) = 3 \Rightarrow \ln(x+1) = \ldots$	M1
	$2 - 4\ln(x+1) = 3 \Rightarrow x = \ldots$ and $-2 + 4\ln(x+1) = 3 \Rightarrow x = \ldots$	dM1
	CVs $\mathrm{e}^{-\frac{1}{4}} - 1$, $\mathrm{e}^{\frac{5}{4}} - 1$	A1
	"-1" $< x < \mathrm{e}^{-\frac{1}{4}} - 1$ or $x > \mathrm{e}^{\frac{5}{4}} - 1$	ddM1A1ft
		(5)
		(9 marks)

10(a)	$\frac{1}{4} = \sin^2 4y \Rightarrow y = \frac{\pi}{24}$	M1A1
		(2)
(b)	$\underline{\underline{\frac{dx}{dy} = 8\sin 4y \cos 4y}}$	<u>M1</u><u>A1</u>
		(2)
(c)	$\frac{dx}{dy} = 8\sin 4y \cos 4y \rightarrow \frac{dy}{dx} = \frac{1}{8\sin 4y \cos 4y}$	M1
	$\frac{dy}{dx} = \frac{1}{8\sqrt{x(1-x)}}$	M1
	$\frac{dy}{dx} = \frac{1}{\sqrt{16 - 64\left(x - \frac{1}{2}\right)^2}}$	A1
		(3)
(d)(i)	$x = \frac{1}{2}$	B1ft
(ii)	$\frac{dy}{dx} = \frac{1}{4}$	B1ft
		(2)
		(9 marks)

(b) Alt	$x = \sin^2 4y \Rightarrow y = \frac{1}{4}\arcsin\sqrt{x} \Rightarrow \frac{dy}{dx} = \frac{1}{4}\frac{1}{\sqrt{1-\left(\sqrt{x}\right)^2}} \times \frac{1}{2}x^{-\frac{1}{2}}$	M1
	$\frac{dx}{dy} = 8\sqrt{x}\sqrt{1-x}$	A1
		(2)
(c)	$\frac{dx}{dy} = 8\sin 4y \cos 4y \rightarrow \frac{dy}{dx} = \frac{1}{8\sin 4y \cos 4y}$	M1
	$\frac{dy}{dx} = \frac{1}{8\sqrt{x(1-x)}}$	M1
	$\frac{dy}{dx} = \frac{1}{\sqrt{16-64\left(x-\frac{1}{2}\right)^2}}$	A1
		(3)